CƠ SỞ DI TRUYỀN HỌC BIỂU SINH

(INTRODUCTION TO EPIGENETICS)

Phuc Loi Luu, PhD

Email: Luu.p.loi@googlemail.com

Zalo: 0901802182

Feb 26 2026

Content

- Class member introduction
- Curriculum of Epigenetics 2026
- Projects and important dates
- Evalutation for the course
- How Epigenetics 2026 lecture work?
- Linux OS, Google Colab, Linux Command Lines and github

Class member introduction

PHUC–LOI LUU, PhD

Current Affiliation	**Head of Scientific Research Office** Institute for Applied Research in Health Sciences and Aging (ARiHA) - Thong Nhat Hospital 01 Ly Thuong Kiet st, Tan Binh District, HCM city, Vietnam
Cơ quan	Viện Nghiên cứu Ứng dụng Khoa học Sức khỏe và Lão hóa Bệnh viện Thống Nhất
Webpage	https://scholar.google.com.au/citations?user=KPIqpJsAAAAJ&hl=en
ORCID	0000-0001-8045-718X
Address **Home**	592/17 Lac Long Quan, P5, Q11, TpHCM 5 Rose st, Sefton, Sydney 2162 Australia
Telephone	+84 (0)901802182 **E–mail** luu.p.loi@googlemail.com loilp@bvtn.org.vn

EDUCATION AND PROFESSION

2023 – 2024	**Tam Anh Research Institute (TAMRI)**	*Head of Data Science Division*	Ho Chi Minh City, Vietnam
2022 – 2023	**Zymo Research**	*Bioinformatic Group Leader*	Ho Chi Minh City, Vietnam
2014 – 2022	**Garvan Institute of Medical Research,** UNSW Sydney	*Bioinformatics researcher in Computational Cancer Genomics and Epigenomics under the supervision of Prof. Susan Clark Conjoint Senior Lecturer*	Sydney, Australia
2011 - 2014	**Max Planck Institute for Molecular Biomedicine,** University of Muenster	*PhD student in stem cell computational biology under the supervision of Prof. Hans R. Schöler*	Muenster, Germany
2010 – 2011	**KIST–Europe**	*Data Scientist*	Saarbruecken, Germany
2008 – 2010	**Max Planck Institute for Informatics,** University of Saarland	*Bioinformatics Post-graduate student under the supervision of Prof. Thomas Langauer*	Saarbruecken, Germany
2005 – 2008	**Nong Lam University** - Ho Chi Minh City	*Lecturer*	Ho Chi Minh City, Vietnam
2000 – 2005	**University of Science** - Ho Chi Minh City National University	*Bioinformatics undergraduate student under the supervision of Prof. Ho Huynh Thuy Duong*	Ho Chi Minh City, Vietnam

Education and Research Interest

Undergraduate at University of Science - Ho Chi Minh City National University (2000 – 2004)

- Major: Bioinformatics
- Research Topics:
  - Genotyping HCV, HBV, HPV
  - Anti-biotics resistance in bacteria

PENICILLIN BINDING PROTEINS (PBPs) IN *STREPTOCOCCUS PNEUMONIAE*: DATABASE SETTING UP AND APPLICATION

Luu Phuc Loi, Thai Ke Quan, Nguyen Hoang Chuong, Thai Thien Minh, Ho Huynh Thuy Duong

Department Genetics, Faculty of Biology, University of Natural Sciences, Viet Nam National University - Ho Chi Minh City

ABSTRACT

β-Lactam antibiotics inhibit the growth of *Streptococcus pneumoniae* by inactivation of cell-wall-synthesizing enzymes, the penicillin-binding proteins (PBPs). Resistance to β-lactam antibiotics of *S. pneumoniae* is essentially due to altered PBPs with decreased affinities to the antibiotic. We investigated relationship between point mutations on pbp genes – pbp1a, 2b, 2x - and penicillin resistance in *S. pneumoniae* clinical isolates. With sequences determined from local isolates combining to data retrieved from GenBank, we established a database named PGD (Penicillin Binding Protein Gene Database) which integrated information about nucleotide sequence and point mutations of 472 strains. Using the set up database, we applied a PCR-based protocol to detect penicillin resistant *S. pneumoniae* strains. The protocol detects 18/20 resistant strains and needs further improvement.

Hepatitis C virus isolate MB14 5' UTR

GenBank: AY690659.1

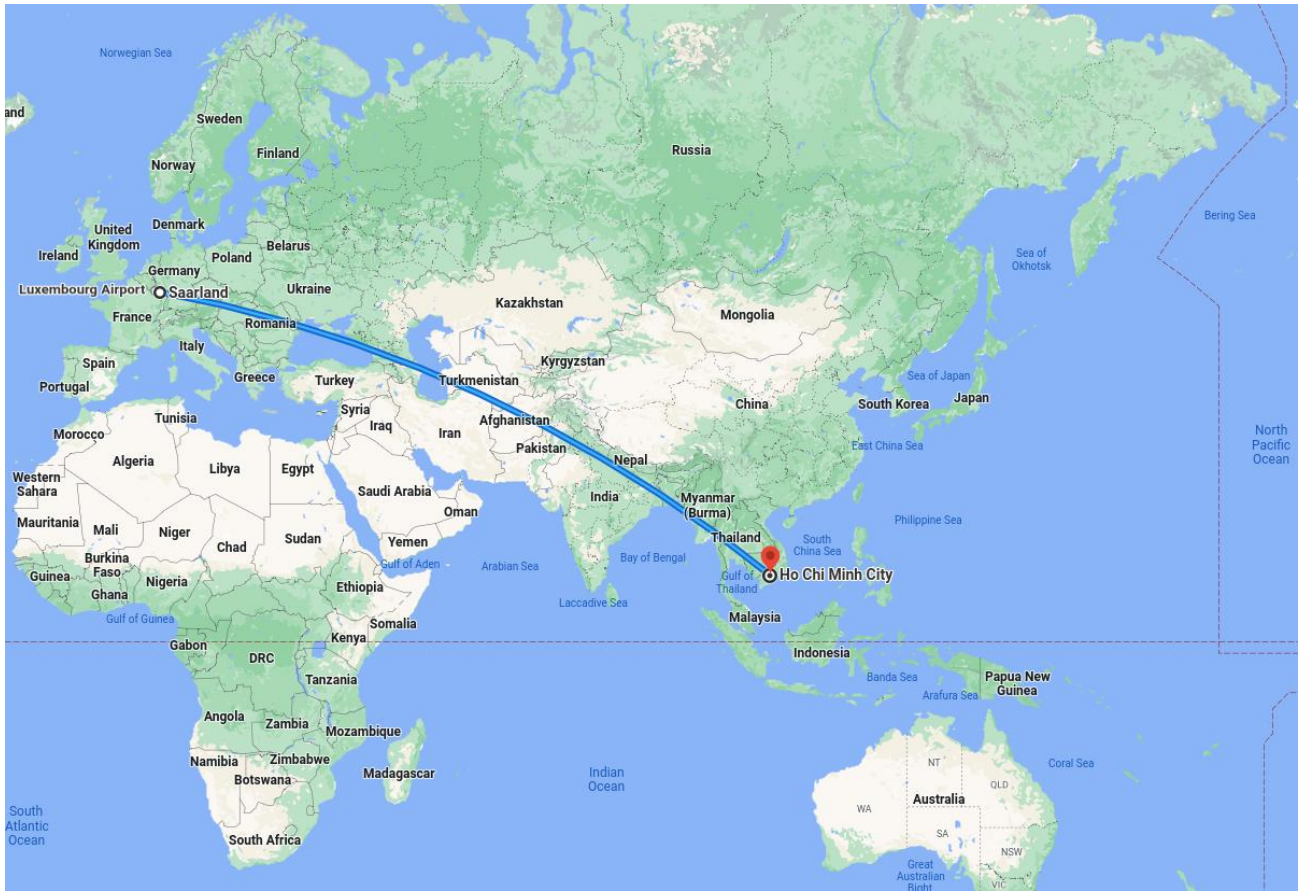
FASTA Graphics PopSet

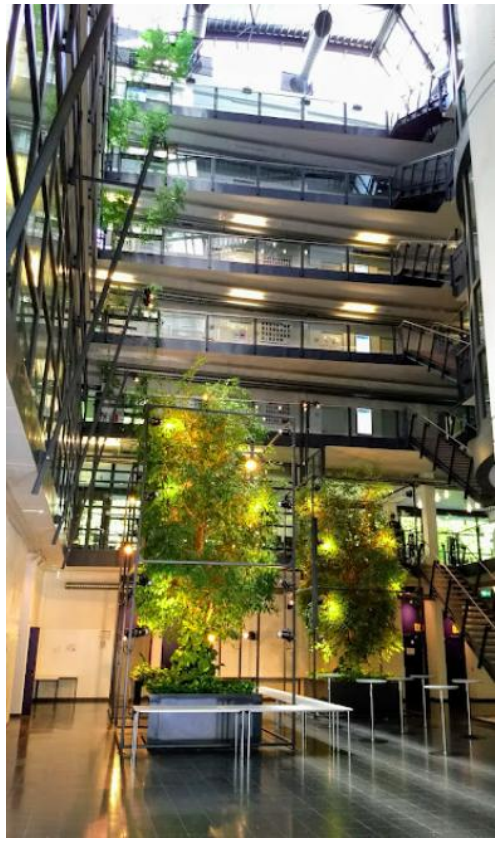
Go to:

```
LOCUS       AY690659                  94 bp    RNA     linear   VRL 22-AUG-2004
DEFINITION  Hepatitis C virus isolate MB14 5' UTR.
ACCESSION   AY690659
VERSION     AY690659.1
KEYWORDS    .
SOURCE      Hepacivirus C
  ORGANISM  Hepacivirus C
            Viruses; Riboviria; Orthornavirae; Kitrinoviricota; Flasuviricetes;
            Amarillovirales; Flaviviridae; Hepacivirus.
REFERENCE   1  (bases 1 to 94)
  AUTHORS   Nguyen,H.C., Ho,T.T.T., Luu,P.L. and Ho,H.T.D.
  TITLE     Genotyping of Hepatitis C virus by real-time RT-PCR
  JOURNAL   Unpublished
REFERENCE   2  (bases 1 to 94)
  AUTHORS   Nguyen,H.C., Ho,T.T.T., Luu,P.L. and Ho,H.T.D.
  TITLE     Direct Submission
  JOURNAL   Submitted (20-JUL-2004) Genetics, University of Natural Sciences
            Viet Nam National University - Ho Chi Minh City, 227 Nguyen Van Cu
            Street Dist. 5, Ho Chi Minh City 0848, Viet Nam
FEATURES             Location/Qualifiers
     source          1..94
                     /organism="Hepacivirus C"
                     /mol_type="genomic RNA"
                     /isolate="MB14"
                     /db_xref="taxon:11103"
                     /country="Viet Nam"
     5'UTR           <1..>94
ORIGIN
        1 ctcgcttcct ggagctgccc ggatccgagg atgatttcat gtgcgatgag ctgcctgacg
       61 gccaggaacg accgggtcct ttcttggatc accc
//
```

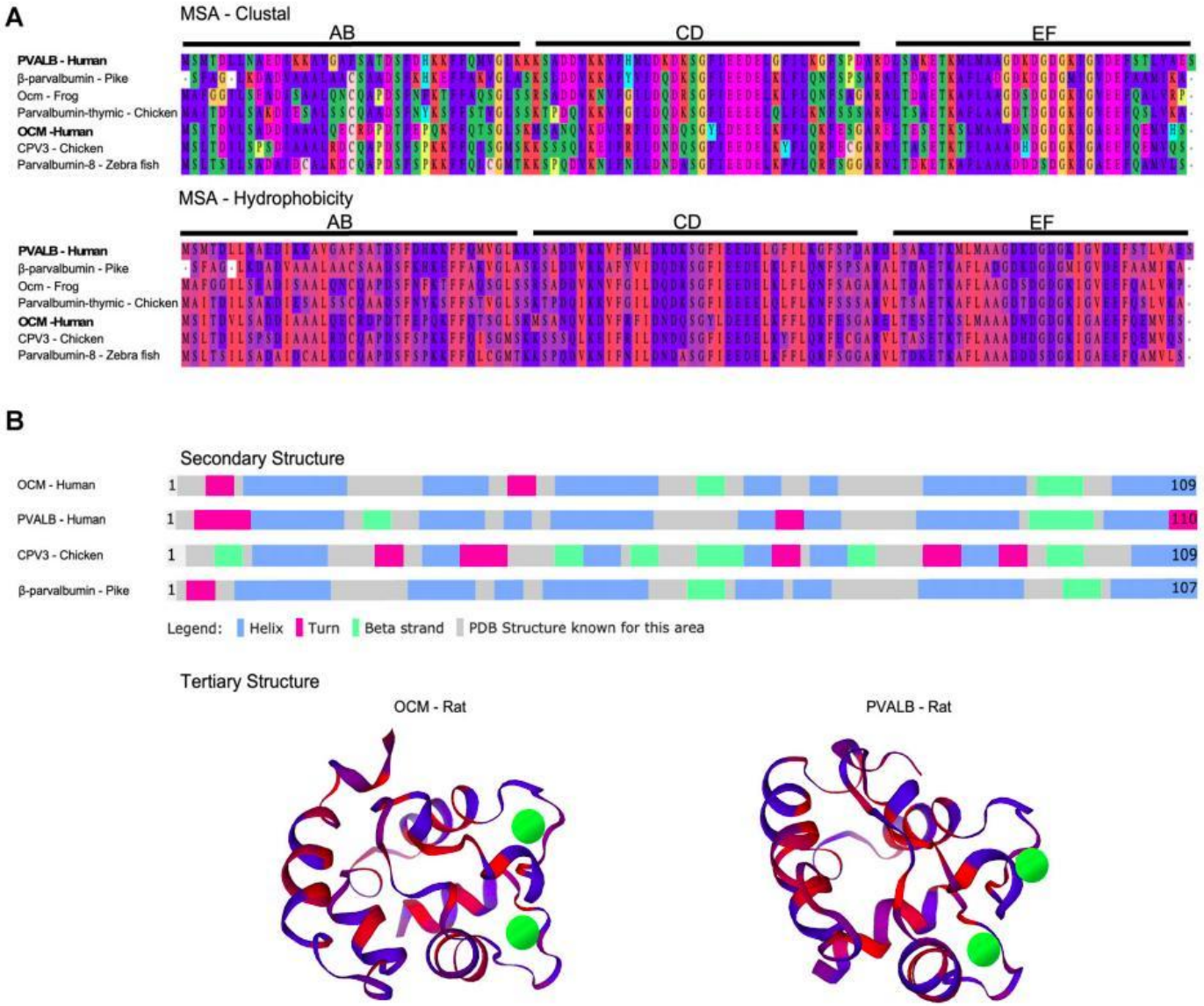
The American journal of tropical medicine and hygiene 73(6):55-55 (2005).

Master degree in **Bioinformatics** at **Max Planck Institute for Informatics**
University of Saarland in Germany **(2007 - 2010)**





Identifying functional discriminative motifs in protein families



- Protein structures and functions
- Bigdata mining and Machine Learning

PhD at **Max Planck Institute for Molecular Biomedicine** and University of Muenster in Germany **(2011 - 2014)**

PhD in Bioinformatics and iPS technology

- Research:
  - The dynamics of stem cell and induced pluripotent stem cell (iPS) epigenomes using genome-wide Next Generation Sequencing (NGS)
  - Somatic memory in iPS technology
  - Bioinformatics pipeline development

CSH PRESS GENOME RESEARCH

HOME | ABOUT | ARCHIVE | SUBMIT | SUBSCRIBE | ADVERTISE | AUTHOR INFO | C

Disclosing the crosstalk among DNA methylation, transcription factors, and histone marks in human pluripotent cells through discovery of DNA methylation motifs

Phuc-Loi Luu[1], Hans R. Schöler[2,3] and Marcos J. Araúzo-Bravo[1,4]

Author Affiliations

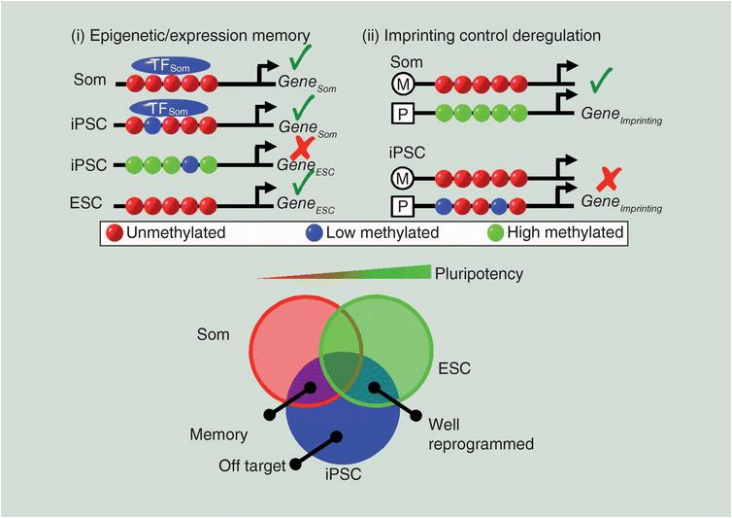
[1]Computational Biology and Bioinformatics Group, Max Planck Institute for Molecular Biomedicine, 48149 Münster, Germany;
[2]Department of Cell and Developmental Biology, Max Planck Institute for Molecular Biomedicine, 48149 Münster, Germany;
[3]University of Münster, Medical Faculty, 48149 Münster, Germany

EPIGENOMICS, VOL. 10, NO. 2 | RESEARCH ARTICLE

Rules governing the mechanism of epigenetic reprogramming memory

Phuc-Loi Luu‡, Daniela Gerovska‡, Hans R Schöler & Marcos J Araúzo-Bravo

Published Online: 16 Jan 2018 | https://doi.org/10.2217/epi-2017-0098



P3BSseq: parallel processing pipeline software for automatic analysis of bisulfite sequencing data FREE

Phuc-Loi Luu, Daniela Gerovska, Mikel Arrospide-Elgarresta, Sugoi Retegi-Carrión, Hans R Schöler, Marcos J Araúzo-Bravo

Bioinformatics, Volume 33, Issue 3, 1 February 2017, Pages 428–431, https://doi.org/10.1093/bioinformatics/btw633

Published: 06 October 2016 **Article history**

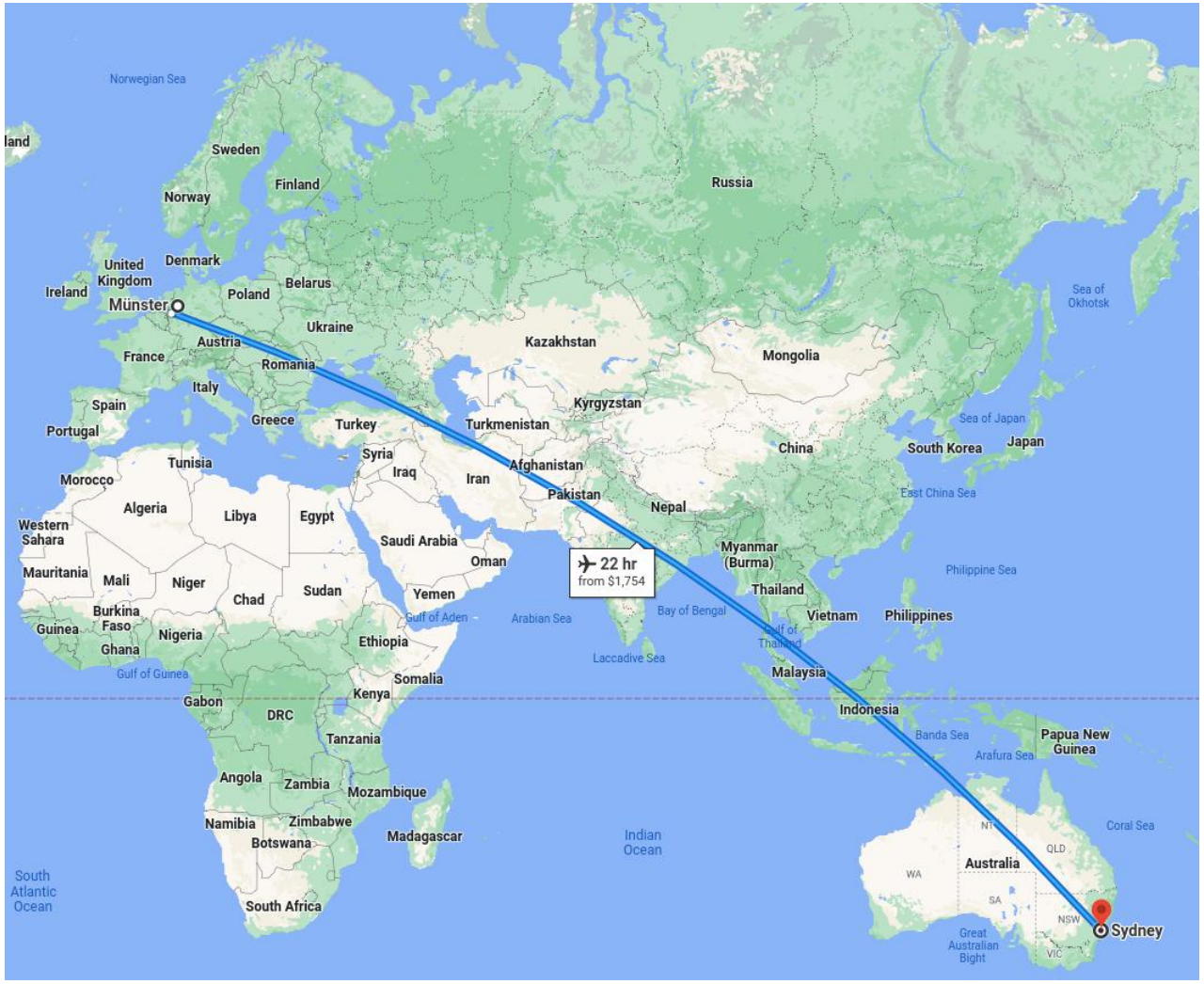
PDF Split View Cite Permissions Share

Abstract

Motivation

Bisulfite sequencing (BSseq) processing is among the most cumbersome next generation sequencing (NGS) applications. Though some BSseq processing tools are available, they are scattered, require puzzling parameters and are running-time and memory-usage demanding.

Seven years of research at Garvan Institute of Medical Research in Australia (2014 - 2022)

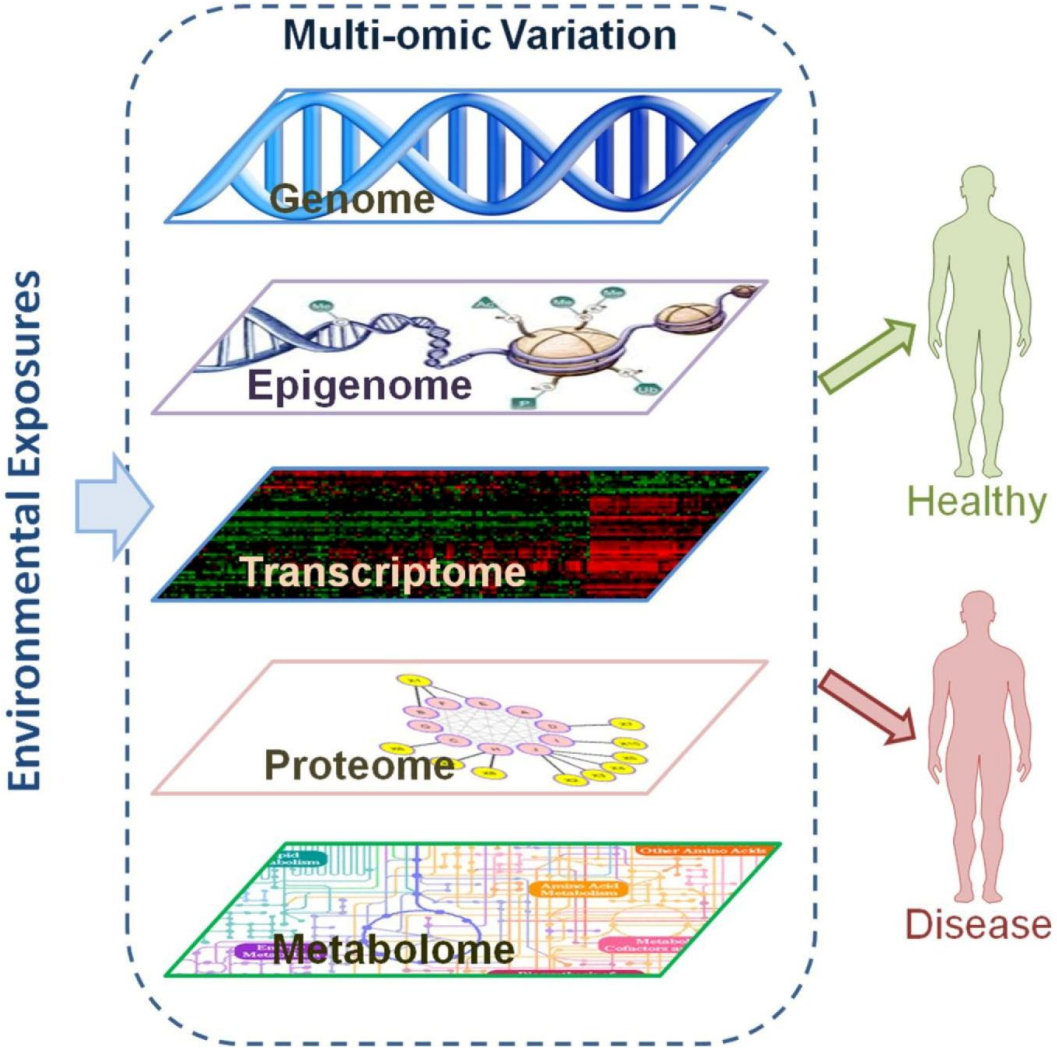


Seven years of research at Garvan Institute of Medical Research in Australia (2014-2022)

- Major: Bioinformatics and Machine Learning
- Research:
  - Cancer Genomics and Epigenomics
  - Single cell and Spatial transcriptomics technology and other diagnosis methods
  - Bioinformatics pipeline development

More than one year at Zymo Research (Sep 2022-Nov 2023)

- Microbiomics
- Epigenomics
- Transcriptomics
- Proteomics
- Metabolomics
- Bioinformatics



Curriculum of Epigenetics 2026

00.0 Introduction to the course "Epigenetics 2025" [Loi] - 01/3/2025
00.1 Epigenetics, Epigenomics and its applications [Loi] - 01/3/2025

01.0 Introduction and overview of Epigenetics [Loi] - 08/3/2025
01.1 Basic Bash script and Awk [Hung/Thanh/Kim] - 08/3/2025

02.0 Chromatin based epigenetic control I [Loi] - 15/3/2025
02.1 Basic programing with R [Hung/Thanh/Kim] - 15/3/2025

03.0 Chromatin based epigenetic control II [Loi] - 22/3/2025
03.1 Statistical analysis with R [Hung/Thanh/Kim] - 22/3/2025

04.0 Histone modifications, TFBSs, DNAaseI Hypersensitivity and ATAC-seq analysis [Loi] - 29/3/2025
04.1 Hands-on: Deeptools, featurecount and DESeq2 [Tam/Minh] - 29/3/2025

05.0 DNA-methylation and epigenetic control [Loi] - 05/4/2025
05.1 Data visualization with R [Hung/Thanh/Kim] - 05/4/2025

06.0 DNA methylation analysis with DMRcate [Thien/Hoang] - 12/4/2025
06.1 DNA methylation visualization with IGV [Hung/Thanh/Kim] - 12/4/2025

Curriculum of Epigenetics 2026

07.0 Non-coding long and small RNAs in epigenetic control [Loi] - 19/4/2025
07.1 Visualizing epigenomics with UCSC Genome Browser [Hung/Thanh/Kim] - 19/4/2025

08.0 Mechanisms of epigenetic control during development [Loi] - 26/4/2025
08.1 Mining Gene Expression GTEX, Epigenome Roadmap and ENCODE [Hung/Thanh/Kim] - 26/4/2025

09.0 Epigenetics and Human Disease Development [Loi] - 03/5/2025
09.1 Mining BLUEPRINT and TCGA database [Minh] - 03/5/2025

10 Mechanisms of epigenetic control in plant I [Nguyen] - 10/5/2025

11 Mechanisms of epigenetic control in plant II [Nguyen] - 17/5/2025

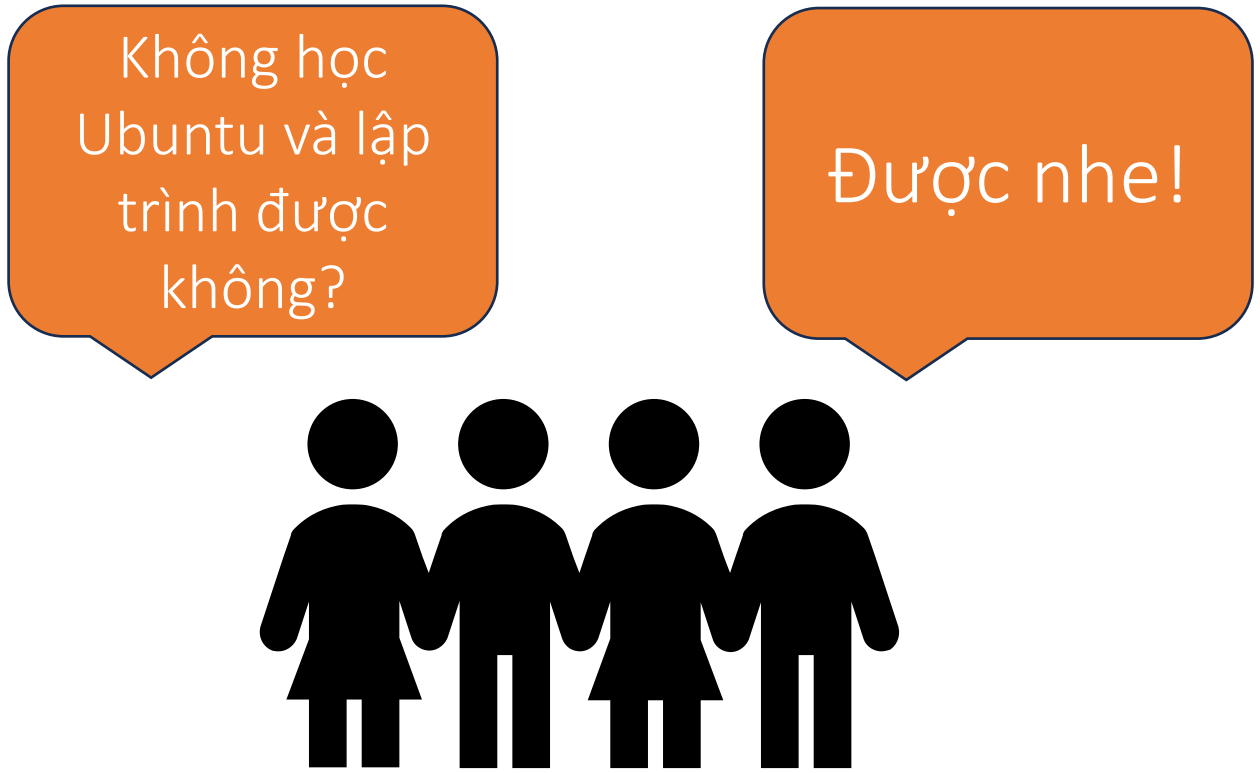
12.0 Epigenetic Biomarkers and epigenetic therapy [Loi] - 24/5/2025
12.1 Course Review [Loi] - 24/5/2025

13.0 Group Project Presenting (31 May 2025) - 31/5/2025

Projects and important dates (**04 June 2026**)

No	Project	Aim	Requirement	Group
1				
2				
3				
4				
5				
6				

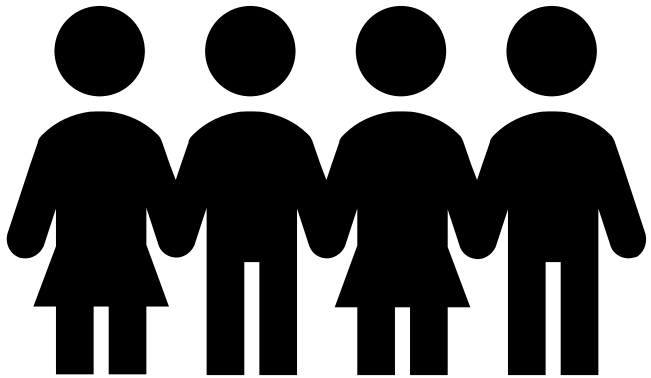
Do we need to install and learn Ubuntu and programing?



Do we need to install and learn Ubuntu and programing?

Không học Ubuntu và lập trình được không?

Được, nhưng như có chân mà ngồi xe lăn!



Evalutation for the course

- Presentation date (final exam): **4 June 2026**
- Max 15 min each group of 2 students
- 5-10 questions each student
- Report (20) +

  Presentation (20) +

  Slide (10) +

  Questions for the presentation (25) +

  Questions for the all of lectures (25)

How does my lecture work?

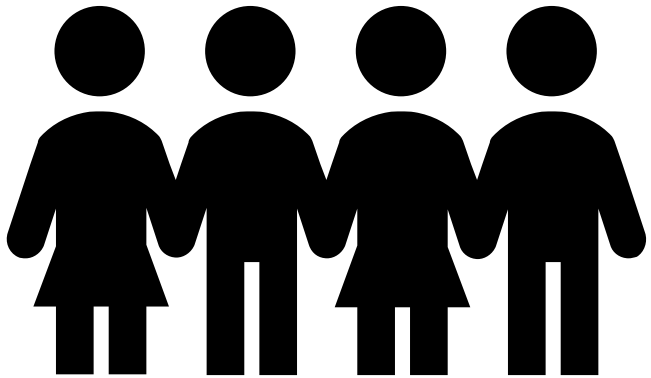
- Start at 18:00 PM every Saturday from **26 Feb 2026** to **4 June 2026**
- 15 min oral test at the begin of the lecture
- Time breaks in a lecture (15 min)
- Lecture end at **21 PM**
- No attendance checking

No attendance checking

No attendance checking

Nghỉ học được không bạn?

Nhưng khó đậu nhe!



https://github.com/luuloi/Epigenetics_2026/

Epigenetics_2026 Public

Pin Watch 0

main 1 Branch 0 Tags Go to file t Add file Code

luuloi Update README.md 4971a82 · now 2 Commits

README.md Update README.md now

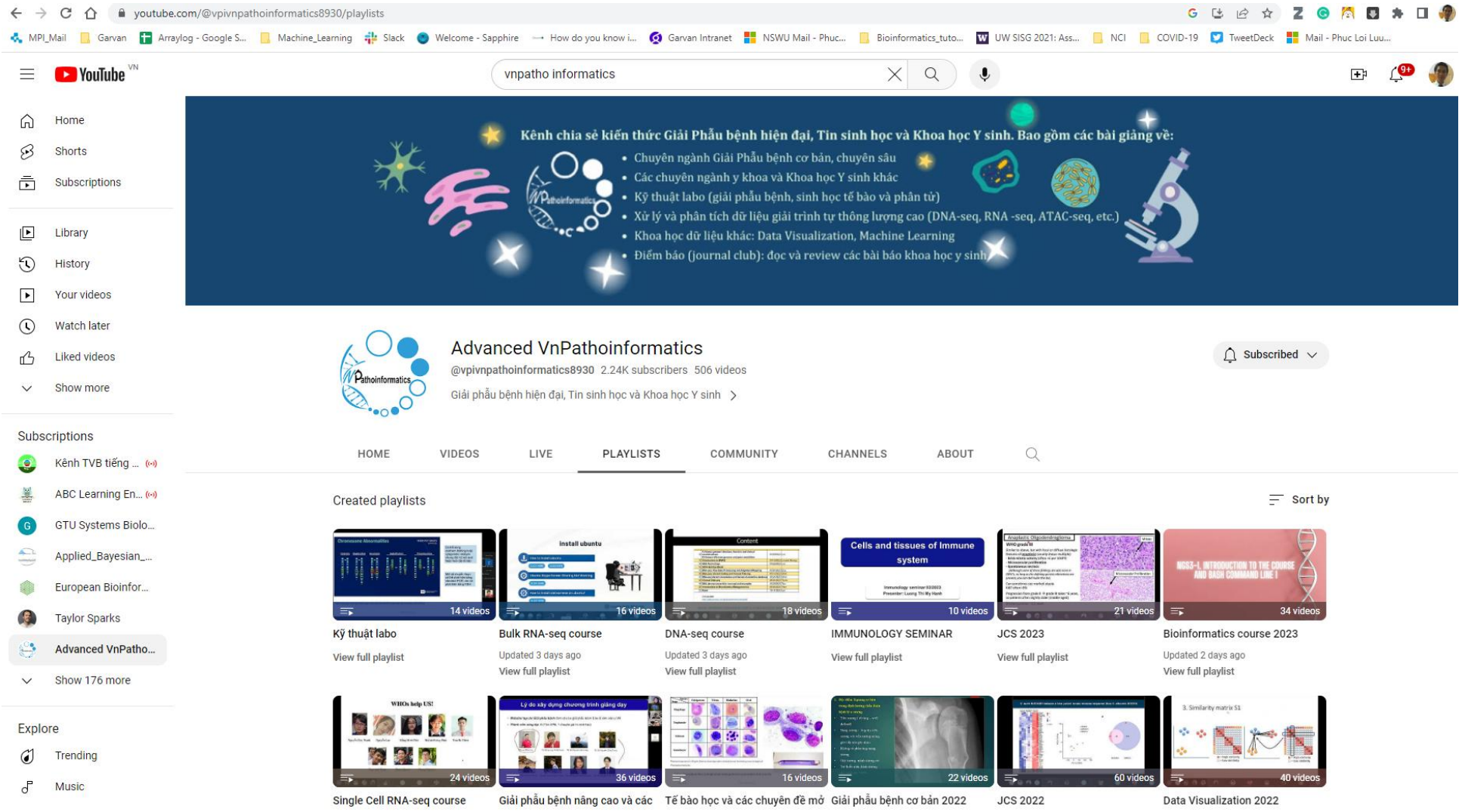
README

Epigenetics 2026

Weekly on Thursday, 18 to 21 PM

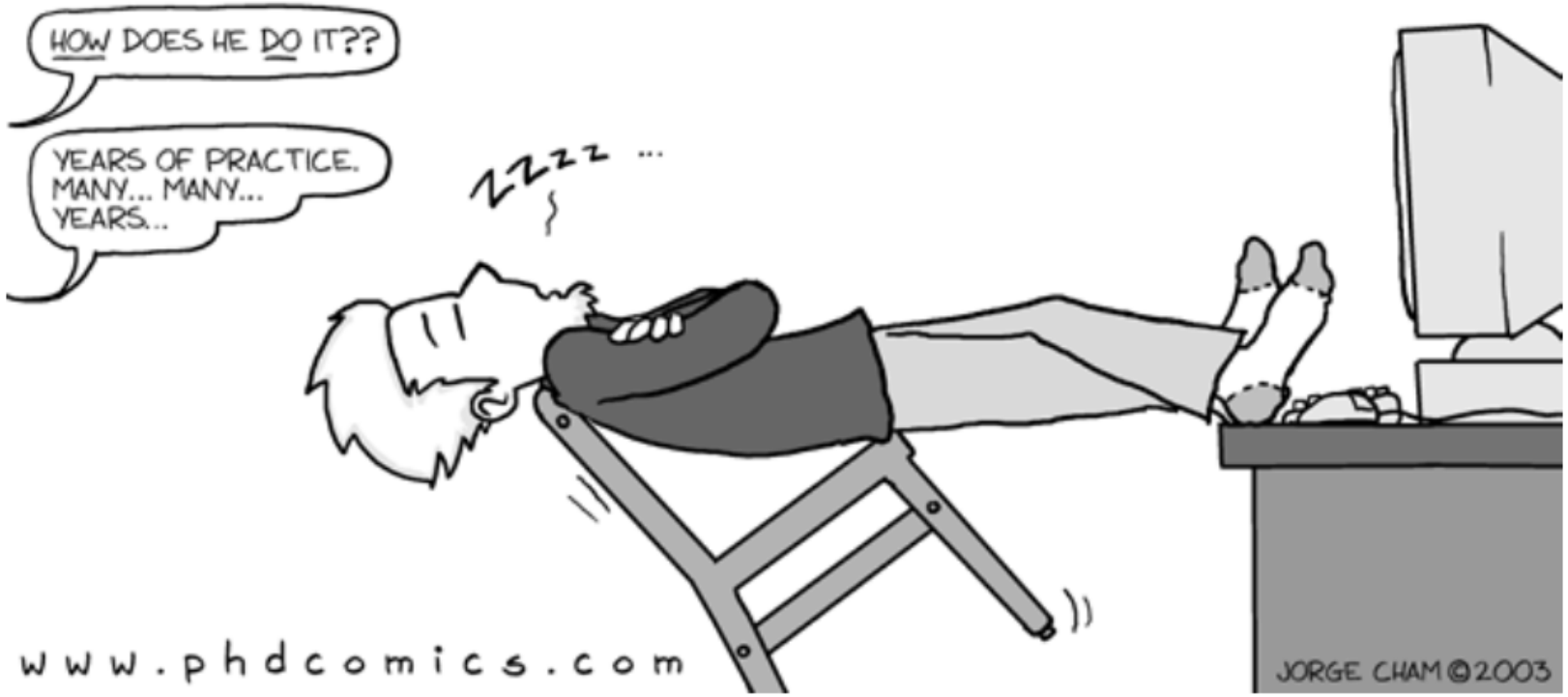
C36, 227 Nguyen Van Cu st, Dist. 5, HCM city

00.0 Introduction to the course "Epigenetics 2026" [Loi] - 26/02/2026
00.1 Epigenetics, Epigenomics and Its Applications [Loi] - 26/02/2026



https://www.youtube.com/@vpivnpathoinformatics8930/playlists

Thank you for your attention!



THANK YOU FOR YOUR ATTENTION!

Please contact:
Email: luu.p.loi@googlemail.com
Zalo: 0901802182
For further information!